

NEXUS Autonomous Remediation Platform

10-Minute Comprehensive Demo Script • Every Function & Underlying Mechanism Explained

TARGET APPLICATION
Farmers Quoting Hub

APM ENGINE
Dynatrace Synthetic

ON-CALL ROSTER
Rugved, Samruddhi, Aarzoo

BUSINESS IMPACT
MTTR: 45m → 3.8m (92% cut)

Timestamp	Functional Area	Detailed Functional Scope & Mechanism
0:00 – 1:30	Stage 0: Overview Cockpit (/)	Explain the SRE Cockpit: 4-stage lifecycle pipeline, 6 microservices health cards, tier classification, SLA tracking, and why traditional NOC monitoring breaks down during storms.
1:30 – 3:00	Stage 1: Dynatrace APM (/monitoring)	Explain the Dynatrace Synthetic probe: HTTP Monitor ID HTTP_MONITOR-06513F5EBB018508, Farmers Intelligent Quoting Hub live URL, 4 core KPI tiles (Status, Availability, N. Virginia location, 357ms performance), and granular network telemetry (DNS, TCP, TLS, TTFB).
3:00 – 4:15	Fault Injection & Correlation (Simulation Bar)	Explain the chaos engine: triggering connection pool starvation. Show the cascade across upstream services and explain how correlation.py compresses 12 raw telemetry alerts into 1 incident (92% noise reduction) with dynamic deduplication keys.
4:15 – 6:00	Stage 2: Gemini AI War Room (/incidents)	Explain the AI diagnostic pipeline: plain-text root cause analysis (sanitized of asterisk markdown), financial impact modeling (,500/min risk), automated runbook mapping (rb-db-pool-recovery), and the 'Simulate Dry Run' engine that verifies DB locks and blast radius safety.
6:00 – 7:30	Stage 3: PagerDuty On-Call (/incident-response)	Explain the incident response dispatch: Tier 1 Primary (Rugved Mhatre via Push & SMS), multi-level escalation policy ladder (Samruddhi Kakade, Aarzoo Sharma), automated ServiceNow CMDB ticket generation (INC0089241), and bi-directional acknowledgment sync.
7:30 – 9:00	Stage 4: Automated Recovery (Remediation Engine)	Explain human-in-the-loop execution: clicking 'Approve & Execute Remediation'. Detail the 3 underlying remediation steps (kill zombie connections, scale pool, health check) and the dual-channel synchronous PagerDuty resolution (Events API v2 + REST API PUT).
9:00 – 10:00	Audit, Postmortem & ROI (/audit & Wrap-Up)	Explain governance & traceability: cryptographically auditable runbook logs, automated postmortem generation, and macro business ROI (MTTR dropped from 45 min to under 4 min, zero manual ticketing).

Act I: SRE Cockpit & Microservices Health Matrix

0:00 – 1:30 (90s)

Interactive On-Screen Flow: Open <http://127.0.0.1:5173/>. Point cursor to: 1) System Health Gauge (100%), 2) The 4-Stage Lifecycle Flow Tracker, and 3) The Microservices Grid below.

What to Say (Verbatim Narrative): "Good morning team. In modern distributed cloud architectures, when an infrastructure bottleneck occurs, operations teams face three critical failures: alert fatigue from hundreds of raw alerts, prolonged Mean Time to Identify while engineers manually read logs, and execution delays while waiting for human runbook approval.

This is **NEXUS**: an Autonomous Incident Remediation Platform that connects **Dynatrace APM**, **Google Gemini AI**, and **PagerDuty** into a self-healing operational loop.

Notice our cockpit layout: across the center is our **4-Stage Lifecycle Flow Tracker**: Deep Observability, AI Correlation & War Room, Incident Response & On-Call, and Automated Runbook Recovery. Below it, our **Microservices Matrix** monitors our 6 core services: Claims Database, API Gateway, Policy Engine, Auth Service, and Payment Processor, tracking real-time latency and SLA adherence."

Under the Hood (Technical Architecture Explanation): Cockpit Architecture: The frontend polls `/api/services` and `/api/analytics` via WebSocket and React state. Each service tile dynamically tracks SLA (99.9% target), response latency in ms, error percentages, and dependencies in the directed acyclic graph (DAG).

Act II: Dynatrace APM Deep Observability & Synthetic Probes

1:30 – 3:00 (90s)

Interactive On-Screen Flow: Click '**Stage 1: Dynatrace APM**' in the sidebar. Highlight the top monitor card: **NEXUS POC - Claims Portal**, click the bracketed link: `([Farmers Intelligent Quoting Hub])` in a new tab, and explain the 4 KPI tiles.

What to Say (Verbatim Narrative): "Let's step into **Stage 1: Deep Observability** powered by **Dynatrace APM**.

Rather than using synthetic mock data, NEXUS is connected directly to an active Dynatrace HTTP Synthetic Monitor: **NEXUS POC - Claims Portal** (Monitor ID: `HTTP_MONITOR-06513F5EBB018508`). Below the title, you can see our direct link to the live production target: the **Farmers Intelligent Quoting Hub**.

Look at our 4 core Dynatrace KPI tiles: **Last Status: Success**, **Availability: 100%**, **Location: 1 probe in N. Virginia** running every 60 seconds, and **Performance: 357 milliseconds** baseline. Beneath these tiles is our granular network breakdown: 14ms DNS lookup, 28ms TCP handshake, 38ms TLS negotiation, and 277ms Time-To-First-Byte (TTFB). Synthetic monitors serve as our critical canary—detecting end-user degradation before users even report it."

Under the Hood (Technical Architecture Explanation): APM Telemetry Pipeline: The backend `dynatrace.py` connector integrates with the Dynatrace Environment API v2. When healthy, probes register 200 OK responses with sub-second latency. When latency exceeds thresholds, Dynatrace emits event anomalies that stream into NEXUS.

Act III: Outage Fault Injection & Event Storm Correlation

3:00 – 4:15 (75s)

Interactive On-Screen Flow: In the top Simulation Bar, click '**DB Connection Pool Exhaustion**'. Show the latency jumping to 4,000+ ms, the status turning CRITICAL, and the raw alert feed filling with 12 alerts.

What to Say (Verbatim Narrative): "Now, let's trigger a realistic enterprise disaster. I will click '**DB Connection Pool Exhaustion**' on our claims database.

Immediately, look at what happens: our Dynatrace Synthetic Monitor fails, latency spikes from 357ms to over **4,200ms**, and cascading timeouts explode across the Claims API, Payment Gateway, and Customer Portal.

In a conventional NOC, this triggers an **Event Storm**: 12 to 15 distinct alerts firing simultaneously across monitoring tools, paging 3 different on-call engineers.

Watch how NEXUS handles this: our event correlation engine (correlation.py) instantly captures all incoming alerts, clusters them by topology dependency and time window, and delivers **92% noise reduction**. Instead of 12 distinct tickets, NEXUS opens exactly ONE correlated master incident linked to the true root cause: **claims-database**."

Under the Hood (Technical Architecture Explanation): Deduplication Engine: correlation.py calculates topological distance using BFS over the microservice graph. Alerts within a 60-second window are correlated into an existing incident using a unique timestamped key (nexus-{service}-{timestamp}), preventing duplicate spam.

Act IV: Gemini AI Incident War Room & Dry-Run Safety Engine

4:15 – 6:00 (105s)

Interactive On-Screen Flow: Click '**Stage 2: Incident Management**'. Open the active CRITICAL incident card. Walk through: 1) Gemini AI Analysis, 2) Business Impact calculation, and 3) Click '**Simulate Dry Run**'.

What to Say (Verbatim Narrative): "Now let's enter **Stage 2: AI Correlation & Incident War Room**.

When an incident triggers, our embedded **Google Gemini AI Copilot** automatically ingests the error logs, topology, and stack traces. Notice the output: it provides a clean, executive-ready explanation of the failure mechanism without clutter or formatting asterisks.

Gemini accomplishes three crucial tasks here:

1. **Root Cause Diagnosis:** It identifies that active connections have reached 100/100 capacity with 45 queries waiting in queue due to unindexed lookups.
2. **Financial & Blast Radius Calculation:** It calculates business impact: 1,420 affected policyholders, 4 critical downstream services degraded, and an estimated revenue risk of **,500 per minute**.
3. **Runbook Recommendation:** It scans our operational runbook registry and pre-stages the exact remediation script: rb-db-pool-recovery.

Before touching production, safety is paramount. Watch as I click '**Simulate Dry Run**'. NEXUS executes non-destructive verification: validating DB read-locks, verifying pool memory overhead, and confirming zero data loss risk before any live execution."

Under the Hood (Technical Architecture Explanation): Gemini Copilot Architecture: gemini_service.py formats prompt context with system metrics, calls the Gemini model, and programmatically strips markdown asterisks for clean rendering. The dry-run engine checks connection limits against maximum safe thresholds.

Act V: PagerDuty Live Mobilization & Bi-Directional Sync

6:00 – 7:30 (90s)

Interactive On-Screen Flow: Click '**Stage 3: Incident Response**' in sidebar. Point out Tier 1 Primary: **Rugved Mhatre**, Escalation Ladder (Samruddhi, Aarzo), and the active P1 incident. Click '**Acknowledge**'.

What to Say (Verbatim Narrative): "While AI was analyzing the problem, how was the human team mobilized? Let's navigate to **Stage 3: Incident Response**.

NEXUS pushed a high-urgency incident to **PagerDuty Events API v2**. Look at our on-call roster: **Rugved Mhatre**, our Tier 1 Core SRE Lead, was paged via mobile push and SMS notification. Standing by on our escalation policy ladder are **Samruddhi Kakade** at Level 2 (+15m) and **Aarzo Sharma** at Level 3 (+30m).

Simultaneously, an enterprise IT ticket—**INC0089241**—was automatically created in **ServiceNow** with full CMDB asset mapping.

Now watch this: when Rugved clicks '**Acknowledge**' on screen (or taps acknowledge on the PagerDuty mobile app), NEXUS bi-directional sync immediately updates the incident status to ACKNOWLEDGED in both NEXUS and PagerDuty within seconds."

Under the Hood (Technical Architecture Explanation): PagerDuty Sync Engine: pd_service.py manages bi-directional synchronization. It maps internal incidents to PagerDuty dedup keys and incident IDs, polling every 15s and syncing acknowledgments and notes directly into PagerDuty's REST API.

Act VI: Automated Runbook Execution & Dual-Channel Auto-Resolve

7:30 – 9:00 (90s)

Interactive On-Screen Flow: Return to Incident Modal (or War Room). Click green button: **'Approve & Execute Remediation'**. Watch the 3-step progress bar execute. Show incident turning to **RESOLVED** and services turning green.

What to Say (Verbatim Narrative): "Now comes the most powerful capability of the platform: **Automated Remediation with Human-in-the-Loop Governance**.

Rather than leaving automation completely unguided, NEXUS requires an authorized SRE to approve execution. I will click **'Approve & Execute Remediation'**.

Watch the 3 automated remediation phases execute live:

1. **Step 1: Drain & Terminate:** Safely kills zombie transactions and idle connections older than 30 seconds.
2. **Step 2: Dynamic Pool Scaling:** Expands max pool size from 100 to 250 connections with zero container restarts.
3. **Step 3: Synthetic Health Verification:** Probes the database endpoint to verify active response times.

Notice the resolution: the incident immediately transitions to **RESOLVED**. And here is our critical integration: NEXUS triggers a **dual-channel synchronous auto-resolution**—sending a resolve event to PagerDuty Events API v2 AND a direct REST API PUT to `/incidents/{id}`. Pagers are silenced, ServiceNow tickets close, and Dynatrace returns to 100% availability."

Under the Hood (Technical Architecture Explanation): Dual-Channel Resolution Engine: Implemented in `remediation_engine.py` and `pd_service.py` (`resolve_incident_everywhere`). It executes OS/database runbook steps, logs execution outputs, and guarantees synchronous PagerDuty resolution across both API channels.

Act VII: Audit Trail, Postmortem Generation & Business ROI

9:00 – 10:00 (60s)

Interactive On-Screen Flow: Click **'Audit Trail'** or return to **Overview Cockpit**. Point to: 1) Cryptographic audit log entry, 2) Zero active outages, and 3) 3.8 min MTTR.

What to Say (Verbatim Narrative): "To conclude our demonstration, let's look at post-incident governance.

In enterprise environments, automation must be 100% auditable. In our **Audit Trail**, every single action—who was paged, when dry-run was simulated, who clicked approve, and the raw terminal outputs of each runbook step—is immutably recorded for compliance.

Furthermore, Gemini AI automatically drafted a standardized postmortem report including timeline, root cause, and follow-up preventative action items.

Let's review the business results:

- **92% Alert Noise Reduction:** 12 noisy alerts compressed to 1 actionable incident.
- **Zero Manual Ticketing Toil:** Bi-directional PagerDuty and ServiceNow synchronization.
- **MTTR Slashed by 91%:** Mean Time to Resolution dropped from a 45-minute manual war room down to **under 4 minutes of automated self-healing**.

Thank you very much. I welcome any questions or technical deep dives."

Under the Hood (Technical Architecture Explanation): Governance & Compliance: All audit events are stored with ISO timestamps and user IDs. The postmortem engine compiles telemetry windows into a Markdown/PDF postmortem ready for SRE retrospective reviews.

■ Executive Presenter Guidelines (10-Minute Pacing):

- **Don't Rush the Explanations:** Spend full time explaining *why* Dynatrace Synthetic monitoring is better than simple ping checks, and *why* dual-channel PagerDuty resolution matters.
- **Highlight Human-in-the-Loop:** Enterprise leaders love AI automation, but fear unguided changes. Emphasize that NEXUS requires human approval and executes safe dry-runs first.
- **Team Personalization:** Pointing out Rugved Mhatre, Samruddhi Kakade, and Aarzo Sharma demonstrates a real, production-ready on-call rota rather than an abstract demo.